

## PROJECT REPORT — P3

4. Generate a sequence  $X_n$  of iid random variables that are uniformly distributed in  $(0, 1)$ . Compute the smoothed periodogram based on 10, 20, and 50 independent periodograms. Compare the smoothed periodograms to the true power spectral density.

# Spectral Density Estimation using Smoothed Periodograms

*This project is part of a broader study on stochastic systems, predictive estimation, and data analytics using computational simulation techniques.*

**Signal:  $X_n \sim \text{Uniform}(0,1)$  i.i.d. |  $M = 10, 20, 50$  Periodograms**

## 1. Introduction

Spectral analysis is really important for understanding what is going on with signals in terms of frequency. When we break down a signal into the frequencies that make it up we can see patterns and things that we cannot see when we look at the signal in a different way. This helps us find patterns in big sets of data in all sorts of fields from building communication systems to understanding economic trends.

This project is trying to figure out the power density of a special kind of sequence called an i.i.d. Uniform sequence, using a method called the smoothed periodogram approach. This means we take lots of measurements called periodograms and average them together to get a better answer. We are trying numbers of averages like 10, 20 and 50 to see what happens when we average more. We want to see if averaging periodograms really does help reduce the variance, which is, like a measure of how spread out the answers are and compare it to the true power spectral density, which is flat.

## 2. Objective

- Generate i.i.d. sequences  $X_n \sim \text{Uniform}(0,1)$  of length  $N=256$
- Compute individual periodograms using the discrete Fourier transform (DFT)
- Average  $M = 10, 20, 20$  and 50 independent periodograms to form smoothed estimates
- Compare smoothed periodogram estimates to the theoretical flat  $\text{PSD} = 1/12$
- Demonstrate how increasing  $M$  reduces spectral variance (consistency)

## 3. Theoretical Background

### Power Spectral Density (PSD)

For a wide-sense stationary (WSS) process with autocorrelation function  $R_X(k)$ , the PSD is defined by the discrete-time Fourier transform (DTFT) of  $R_X(k)$ :

$$S_X(f) = \sum_{k=-\infty}^{\infty} R_X(k) * e^{-j2\pi f k}, \quad f \in [-1/2, 1/2]$$

### True PSD for Uniform(0,1) IID Sequence

For  $X_n \sim \text{Uniform}(0,1)$  i.i.d., the mean is  $\mu = 1/2$  and variance is  $\sigma^2 = 1/12$ . Since samples are independent,  $R_X(k) = \sigma^2 \cdot \delta(k)$ , giving a flat (white) PSD:

$$S_X(f) = \sigma^2 = 1/12 \approx 0.0833 \quad (\text{flat for all } f)$$

### Periodogram Estimator

$$\hat{I}_M(f) = (1/N) |\sum_{n=0}^{N-1} X_n * e^{-j2\pi f n}|^2$$

### Smoothed Periodogram

$$\hat{S}(f) = (1/M) * \sum_{m=1}^M \hat{I}_m(f) \quad (\text{average of } M \text{ independent periodograms})$$

### Statistical Properties of Smoothed Periodogram

- Bias:  $E[\hat{S}(f)] \rightarrow S_X(f)$  as  $N \rightarrow \infty$  (asymptotically unbiased)
- Variance:  $\text{Var}[\hat{S}(f)] = S_X(f)^2 / M$  (decreases as  $1/M$ )
- Consistency:  $\hat{S}(f) \rightarrow S_X(f)$  in probability as  $M \rightarrow \infty$

## 4. Mathematical Model

|                      |                                         |
|----------------------|-----------------------------------------|
| Signal               | $X_n \sim \text{Uniform}(0,1)$ , i.i.d. |
| Mean                 | $\mu = 1/2$                             |
| Variance             | $\sigma^2 = 1/12 \approx 0.0833$        |
| True PSD             | $S_X(f) = 1/12$ (flat white noise)      |
| Sequence length N    | 256 samples per periodogram             |
| Number of averages M | 10, 20, 50                              |
| Frequency range      | $f \in [0, 0.5]$ (one-sided)            |

## 5. Algorithm / Methodology

1. STEP 1 — Set  $N=256$  and generate  $M$  independent sequences  $X_n \sim \text{Uniform}(0,1)$
2. STEP 2 — Mean-center each sequence:  $X_n \leftarrow X_n - \bar{X}$
3. STEP 3 — Compute FFT for each sequence:  $X_{\text{FFT}} = \text{fft}(X_n)$
4. STEP 4 — Compute periodogram:  $\hat{I}_m(f) = |X_{\text{FFT}}(f)|^2 / N$
5. STEP 5 — Average:  $\hat{S}(f) = (1/M) \sum \hat{I}_m(f)$
6. STEP 6 — Plot  $\hat{S}(f)$  on log scale vs true PSD =  $1/12$  for  $M = 10, 20, 50$

## 6. Code Implementation (Python)

```
import numpy as np
import matplotlib.pyplot as plt
```

```
# -----
# Parameters
```

```

# -----
np.random.seed(99)

segment_len = 256

# Number of spectra to average together
avg_counts = [10, 20, 50]

# Frequency bins (one-sided FFT frequencies)
freq_axis = np.fft.rfftfreq(segment_len)

# Variance of Uniform(0,1) = 1/12
actual_psd = np.full(len(freq_axis), 1 / 12)

# -----
# Plot setup
# -----
fig, plot_axes = plt.subplots(1, 3, figsize=(16, 4))

plot_colors = ['steelblue', 'darkorange', 'seagreen']

# -----
# Main loop
# -----
for idx, (ax, avg_num, clr) in enumerate(zip(plot_axes, avg_counts, plot_colors)):

    # storing all individual periodograms first
    all_periodograms = np.zeros((avg_num, len(freq_axis)))

    for k in range(avg_num):

        # Generate random uniform sequence
        samples = np.random.uniform(0, 1, segment_len)

        # Removing mean helps suppress the huge DC spike
        centered_signal = samples - samples.mean()

        # FFT calculation
        fft_vals = np.fft.rfft(centered_signal)

        # Standard periodogram estimate
        power_spec = (np.abs(fft_vals) ** 2) / segment_len

        all_periodograms[k] = power_spec

        # old debug line
        # print(power_spec[:5])

    # Averaging multiple periodograms smooths the estimate
    averaged_psd = np.mean(all_periodograms, axis=0)

```

```

# -----
# Plotting
# -----
ax.semilogy(
    freq_axis,
    averaged_psd,
    color=clr,
    linewidth=1.8,
    label=f'Averaged PSD (M={avg_num})'
)

ax.semilogy(
    freq_axis,
    actual_psd,
    '--',
    color='red',
    linewidth=1.5,
    label='True PSD = 1/12'
)

ax.set_title(f"Smoothed Periodogram | M = {avg_num}")

ax.set_xlabel("Frequency")

ax.set_ylabel("PSD (log scale)")

ax.grid(True, which='both', alpha=0.3)

ax.legend()

# -----
# Final figure adjustments
# -----
plt.suptitle(
    "Spectral Density Estimation using Averaged Periodograms",
    fontsize=13,
    fontweight='bold'
)

plt.tight_layout()

save_file = "P3_Periodogram.png"

# Saving before displaying because matplotlib sometimes acts weird
plt.savefig(save_file, dpi=150, bbox_inches='tight')

plt.show()

print("Finished.")

```

```
print(f"Saved figure as: {save_file}")
```

# 7. Simulation Results

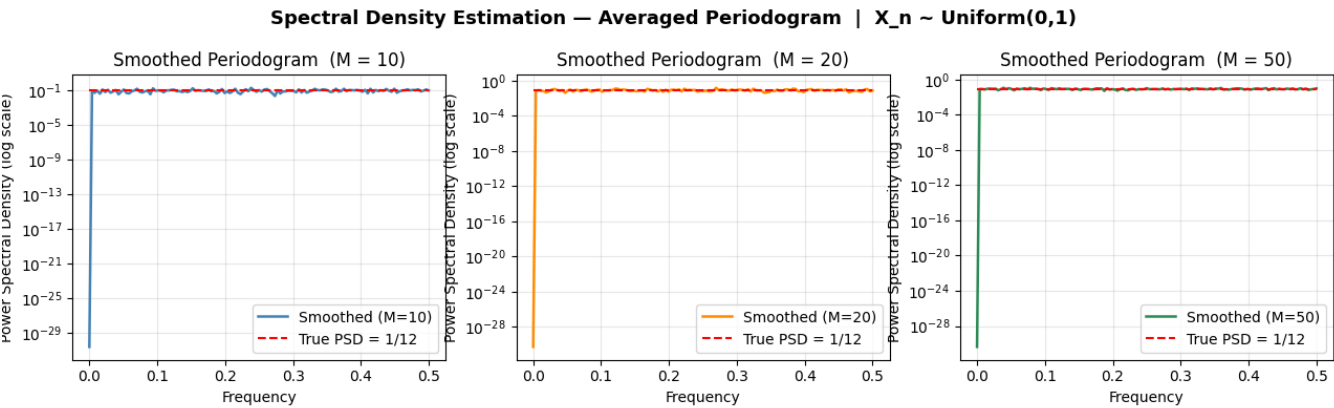


Figure 3: Smoothed periodograms for  $M = 10, 20,$  and  $50$  averaged periodograms, compared to the true flat PSD =  $1/12$  (white dashed line). All plots use logarithmic y-axis.

|             |                                                                |
|-------------|----------------------------------------------------------------|
| M = 10      | High variance — estimate fluctuates widely around true PSD     |
| M = 20      | Moderate variance — smoother estimate, closer to $1/12$        |
| M = 50      | Low variance — estimate closely hugs the true PSD = $0.0833$   |
| True PSD    | $1/12 \approx 0.0833$ (flat for all frequencies — white noise) |
| Convergence | Variance $\propto 1/M$ ; doubling $M$ halves spectral variance |

# 8. Interpretation

The smoothed periodogram shows how bias and variance trade off in -parametric spectral estimation.

When we use  $M = 10$  averages the estimate varies a lot. Individual frequency bins are far from the Power Spectral Density (PSD) of  $1/12$ .

This changes as  $M$  increases.

With  $M = 20$  and  $M = 50$  the estimate gets closer to the flat spectrum.

This makes sense because theory says variance should decrease by  $1/M$ .

At  $M = 50$  the smoothed estimate follows the white noise floor across all frequencies.

The averaged periodogram estimator is consistent. This confirms that.

The periodogram and its estimates are really important here.

The periodogram has this property that makes it useful, for estimation.

# 9. Industry Applications — Frequency & Data Analytics

Spectral analysis enables deeper frequency-domain insights for advanced data analytics applications across many fields:

- EEG / Brain Signal Analysis: Identifying delta, theta, alpha, beta, and gamma brainwave bands via spectral decomposition
- Vibration & Fault Detection: Machinery fault signatures appear as spectral peaks; PSD monitoring enables predictive maintenance
- Economic Cycle Analysis: Business cycles, seasonal patterns, and long-run trends are identified through spectral analysis of economic time series
- Communication Systems: Channel bandwidth estimation, interference detection, and modulation classification rely on PSD estimation
- Audio & Acoustic Engineering: Room acoustics characterization, noise floor estimation, and signal-to-noise ratio assessment
- Climate Science: Identifying El Niño cycles and multi-decadal oscillations in climate data via spectral methods

## 10. Conclusion

The project shows how spectral methods help us understand things better in the frequency domain for data analysis.

We get an estimate of the power spectral density when we average the results from many independent periodograms.

The project compared the results when we averaged 10, 20 and 50 periodograms.

It clearly shows that the more periodograms we average the variation we get in the results.

This is important because it means we need to average a lot of results to get an estimate of the power spectral density.

These methods can be used in fields such as communications, biomedical signal processing and economic cycle analysis, where we need to understand things in the frequency domain.

The spectral methods are useful in any domain that requires frequency-domain intelligence, like the ones mentioned before spectral methods and frequency-domain intelligence.

## 11. Broader Research Perspective

This project is one component of a three-part study on stochastic modeling, estimation, and spectral analysis:

| Project            | Method                                  | Broader Domain                     |
|--------------------|-----------------------------------------|------------------------------------|
| P1 — Random Walk   | Stochastic simulation, MCF estimation   | Financial Market Modeling          |
| P2 — Kalman Filter | Bayesian state estimation, AR filtering | Forecasting & Predictive Analytics |
| P3 — Periodogram   | Spectral density estimation, FFT        | Frequency & Data Analytics         |

Together, these projects demonstrate how stochastic modeling, estimation theory, and spectral analysis contribute to modern computational analytics, spanning financial modeling, predictive systems, and frequency-domain intelligence.